

HACK YOUR PATH 7.0

Theme : Open Innovation (SDG-Alligned)

Problem Statement Title - CortexKey:
Brainwave-Backed Authentication

Team Name : BlackHats

Team Details :

1. Devesh CSD 2nd yr
2. Aditya CSD 2nd yr
3. Sadaf CSE 1st yr
4. Abhinav CSE 1st yr

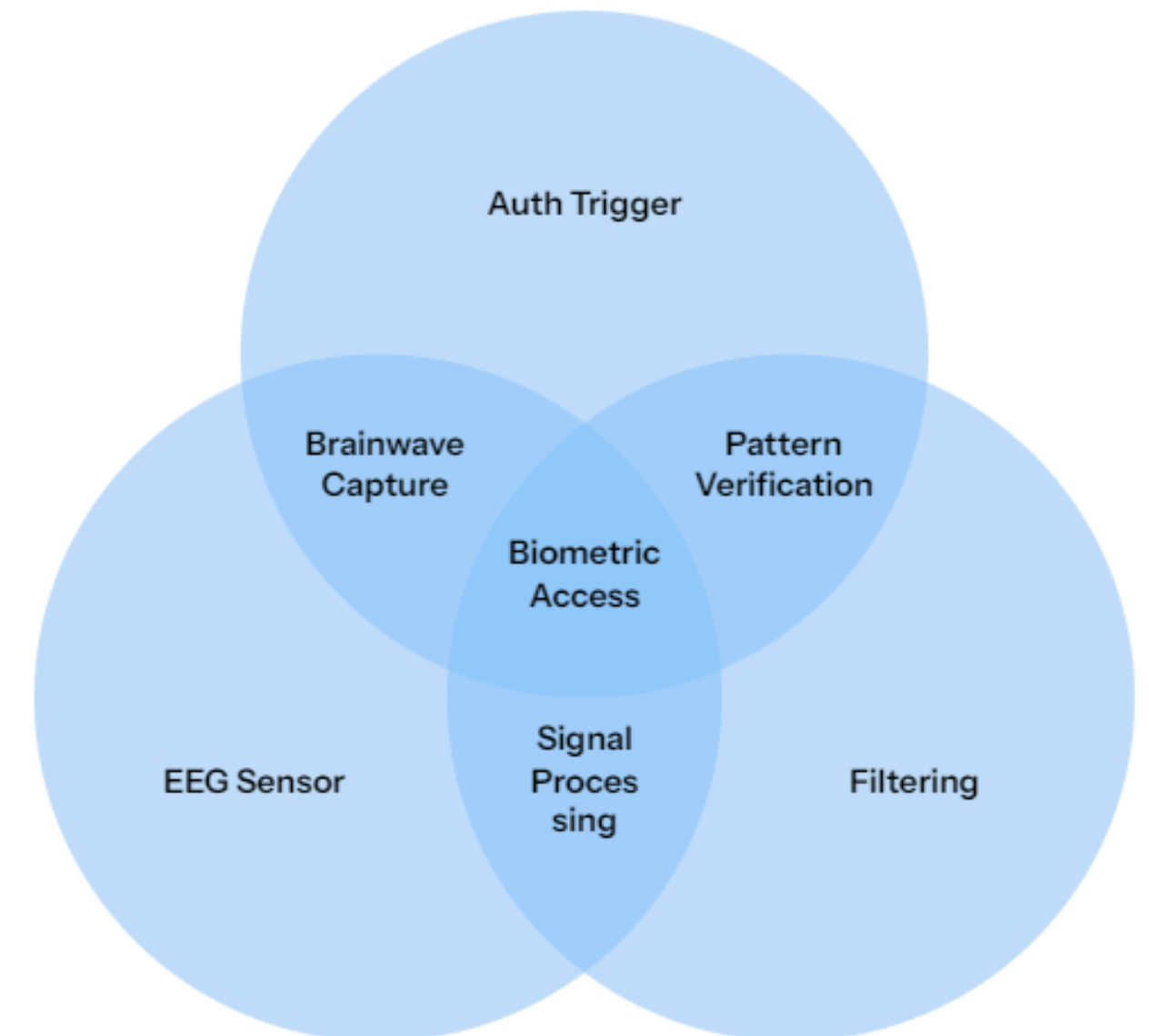


IDEA TITLE

Problem Statement

Current authentication methods fall short:

- Passwords are reusable, guessable, and vulnerable to phishing
- Biometrics like Fingerprint and FaceID rely on static physical traits
- Once compromised, biometric data cannot be changed
- Single-factor authentication increases the risk of unauthorized access



Solution: Evolution of Authentication Systems

From static credentials to brainwave-based biometric login

CortexKey – Secure Login Using Brainwaves

- Traditional biometrics authenticate using physical traits
- These can be replicated or misused

CortexKey:

- Uses EEG brainwave features for login authentication
- Acts as an add-on or alternative to FaceID/fingerprint
- Ensures strong liveness-based identity verification



**SUSTAINABLE
DEVELOPMENT
GOALS**



Future of Authentication

CortexKey is poised at the forefront of next-gen security, evolving with wearable tech.

Today

Proof of concept, rapid prototyping, and foundational architecture.

Near Term

Pilot programs in high-security environments and educational platforms.

Long Term

Integration into mainstream consumer BCI wearables and ubiquitous secure access.

TECHNICAL APPROACH



Hardware Layer

Leveraging cutting-edge bio-sensing technology for accurate signal capture.

- **Sensor:** Neuphony EXG Synapse or BioAmp EXG Pill (Analog Front-End)
- **MCU:** ESP32 or Arduino Uno R3 (Data transmission)
- **Power:** Isolated DC source to minimize 50Hz mains noise

Software Layer

Efficient and reliable code for seamless data handling and system control.

- **Languages:** Python 3.8+ (Backend), C++ (Firmware)
- **Drivers:** Standard Serial/UART communication drivers

Processing Pipeline

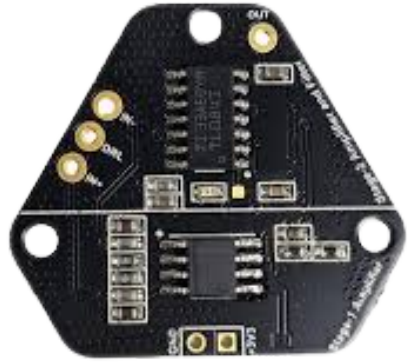
A four-stage process transforming raw brainwaves into an authentication trigger.

1. **Acquisition:** Raw capture via Frontal Lobe electrodes.
2. **Preprocessing:** Digital 50Hz Notch Filter & 5Hz–30Hz Bandpass Filter.
3. **Extraction:** Power Spectral Density (PSD) analysis.
4. **Authentication:** Lightweight SVM Classifier matching.

ML & Libraries

Utilizing industry-standard tools for signal processing and machine learning.

- **Signal Processing:** MNE-Python, SciPy
- **Machine Learning:** Scikit-learn (SVM/Decision Tree)



FEASIBILITY AND VIABILITY

Feasibility:

- Uses existing EEG signal processing methods
- Low-cost, non-medical EEG hardware
- Software-driven authentication logic
- Prototype buildable within hackathon time
- No dependency on large ML models

Viability

- Addresses real security gap in passkeys & biometrics
- Works as a second factor, not a replacement
- Scales with improving wearable EEG tech
- High relevance for secure systems & exams
- Strong future adoption in high-security domains

Item	Product	Approx. Price (INR)
BCI Sensor	Neuphony EXG Synapse OR BioAmp EXG Pill	₹1,200 – ₹1,500
Microcontroller	Arduino Uno R3 (Clone) or ESP32	₹450 – ₹600
Electrodes	Gel-based disposable ECG/EEG studs	₹200 – ₹300
Cables	Bio-potential cables (Snap-on type)	₹300 – ₹400
Power	9V Battery	₹100
Total		~₹2,500 – ₹2,900

NOTE:

The price range for a standard hardware passkey falls between ₹3,000 and ₹20,000.

RESEARCH AND REFERENCES

Academic Publications

- [Chuang, J., Nguyen, H., & Sayed, B. \(2013\).](#) "Cerebrem: A Brain-Computer Interface for User Authentication." Demonstrates the use of mental tasks (e.g., imagining movement) as a viable, secure cryptographic key.
- [Gui, Q., Zhan, Z., & Zhu, J. \(2014\).](#) "A Survey of EEG-based Biometrics for User Authentication." Validates brainwaves as high-entropy, non-static biometric identifiers superior to fingerprints.
- [Ashby, C., et al. \(2011\).](#) "Low-Cost EEG Based BCI User Identification." Proves that non-medical grade EEG sensors (like the EXG Pill) are sufficient for robust identity verification.
- [Palaniappan, R. \(2004\).](#) "Method of Identifying Individuals Using Brain Electrical Activity Features." The foundational study for using Power Spectral Density (PSD) as a "Neural Signature."

Technical Frameworks & Tools

- **BrainFlow SDK:** The industry-standard API for real-time biosensor data acquisition across open-source hardware.
- **MNE-Python:** Primary library for scientific-grade EEG signal processing, noise filtering, and artifact removal.

PARTICIPATION IN OTHER HACKATHONS (AUG 25 to JAN 26)

Blackhats

Team Member	Hackathon Details (Aug 25 to Jan 26)	Achievement if any
Devesh	Internal SIH25	2nd Prize
Aditya	Internal SIH25	2nd Prize
	NXT-WAVE Buildathon	qualified for round 2
Sadaf	Methodist 24hrs hackathon	1st Prize out of 45 teams
	AI for bharat	Ongoing
Abhinav	Internal SIH25	Summited in portal
	Frost labs HITAM	None